EFY TESTED
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SMART ELECTRONIC VOTING MACHINE (EVM)

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The motivation behind this project is to create an electronic voting machine (EVM) that will help overcome the challenges posed by manual voting machines. It aims to be cost-effective, time-saving, and eliminate potential fraudulent activities within the paper-based voting system. In scenarios such as village elections, school elections, or college elections, this smart and budget-friendly EVM can ensure fair and efficient electoral processes.

The presented project features eight switches that are labelled S1 to S8, each corresponding to a candidate, and additional switch S9 that is designated for displaying the results. An LED is connected to D3 (pin 18) of Arduino Uno, while a buzzer is connected to D2 (pin 17) to indicate vote status. Voters can cast their votes for their preferred candidates. The cumulative vote count for each candidate is displayed on an LCD screen, and the final result can be calculated by pressing the result button (switch S9). Fig. 1 shows the author's prototype.

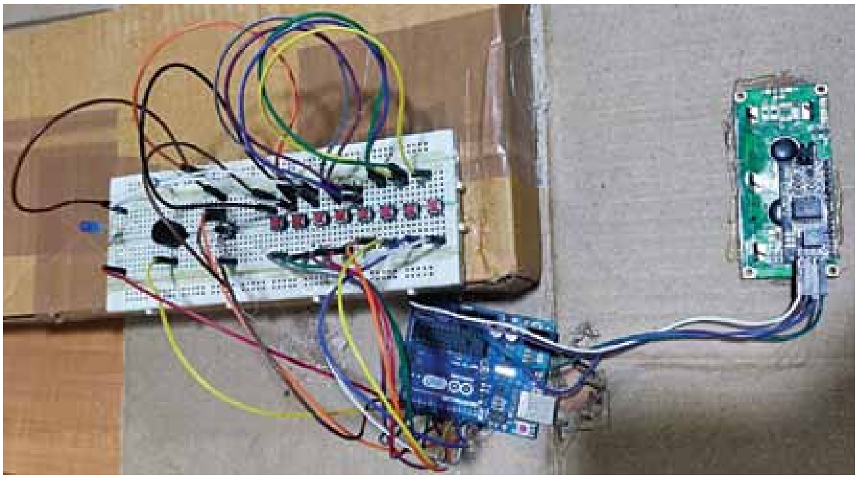

Fig. 1: The author's prototype

Circuit and working

The circuit diagram for the EVM project is depicted in Fig. 2. The design is based on Arduino Uno and is centred around components such as an I2C LCD (module1), Arduino Uno (board1), nine pushbutton switches (S1 through S9), jumper wires, a breadboard, and various other components.

Switches S1 to S8 enable voters to cast their votes for the eight candidates, while switch S9 facilitates the calculation of results, which are then displayed on the LCD.

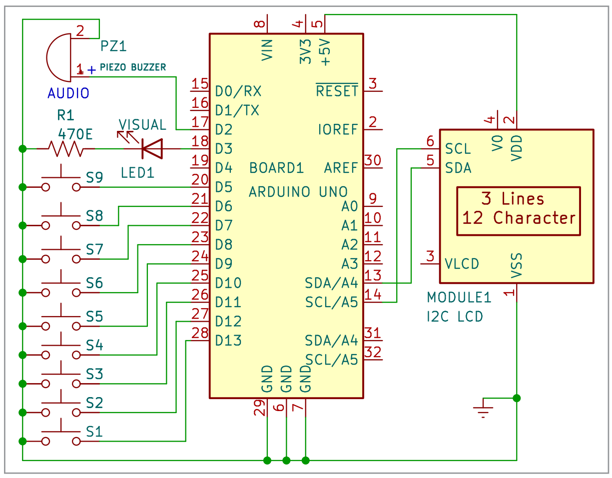

Fig. 2: Circuit diagram of the smart EVM using Arduino Uno

The Arduino Uno (board1) is a microcontroller board based on ATmega328P processor, featuring 14 digital input/output pins (with 6 PWM outputs), 6 analogue inputs, a 16MHz resonator, USB connectivity, a power jack, an ICSP header, and a reset button.

An I2C LCD (module1) display primarily consists of an HD44780-based character LCD display and an I2C LCD adaptor. These LCDs are designed for character-based displays and can show 32 ASCII characters across two rows, such as the 16 × 2 character LCD. The adaptor employs 8-bit I/O expander chip PCF8574, which converts I2C data from the Arduino into parallel data suitable for the LCD display.

The piezo buzzer (PZ1) produces sound when a voter casts a vote by pressing the corresponding pushbut-